

Georgy Savva

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RESEARCH INTERESTS

World Models, Representation Learning, Multimodal Foundational Models, Agent Learning

EDUCATION

New York University <i>Master of Science in Computer Science, Courant Institute of Mathematical Sciences. GPA: 3.8</i>	New York, United States 2024 – 2025
Russian University of Cooperation <i>Bachelor of Science in Computer Science</i>	Moscow, Russia 2019 – 2021
National Research Nuclear University MEPhI <i>Bachelor of Science in Cybersecurity</i>	Moscow, Russia 2014 – 2016

EXPERIENCE

Member of Technical Staff, Research <i>Runway</i> Exploring intuitive physics understanding in foundational video models	Jun 2026 – Present New York, United States
Research Assistant <i>NYU Courant Institute of Mathematical Sciences, Advised by Prof. <u>Saining Xie</u></i> - Developed the first multi-agent Minecraft World Model - Investigated quality degradation in video world models by applying Diffusion Forcing and Self Forcing	Jan 2025 – Feb 2026 New York, United States
Research Intern <i>NYU Courant Institute of Mathematical Sciences, Advised by Prof. <u>Lerrel Pinto</u></i> - Applied online reinforcement learning to fix the human demonstration morphology gap for dexterous robotic manipulation - Trained a behavior cloning policy using transformers on third-person human demonstration data to solve dexterous robotic hand tasks - Set up an auto-resettable reinforcement learning environment for online robot training using the spacemouse device	Jun 2024 – Nov 2024 New York, United States
Senior Software Engineer <i>Raylu</i> - Led backend development of an LLM chatbot for healthcare and deployed it to production, serving first paying customers - Developed an LLM-powered SaaS workflow automation application using the open source workflow engine Activepieces, allowing the company to release a new product in under 3 months	Oct 2023 – May 2024 New York, United States
Technical Co-Founder <i>Scifind</i> - Launched a troubleshooting platform for bioscientists into production, gaining 4,000 MAU in the first 2 months - Led the development of the product with a team of 2 engineers using TypeScript, Next.js, and Node.js	Jul 2022 – Aug 2023 Los Angeles, United States
Senior Software Engineer <i>IOTA</i> - Developed a new network layer of the blockchain node using libp2p, decreasing the number of peering errors by 70% - Set up automatic deployment via Ansible and GitHub Actions, accelerating the development speed by 25% - Introduced Go guidelines to the project and tools to ensure them, reducing the frequency of bugs by 50%	Jan 2021 – Jul 2022 Berlin, Germany
Senior Software Engineer <i>Elsa</i> Implemented App Store and Google Play in-app subscriptions as microservices using FastAPI, which improved the correctness of billing by 50%	Jan 2020 – Jan 2021 San Francisco, United States

- Integrated payment gateway Instamojo, increasing the number of app purchases in Asia threefold

Software Engineer

Mar 2017 – May 2019

Edwin

San Francisco, United States

- Participated in building the English tutor chatbot from a prototype stage to being the top 1 bot on Facebook
- Implemented a YAML-file-based framework for building dialog systems, which allowed the company to launch a new product on Google Assistant
- Designed a distributed queue using PostgreSQL to handle messages from 800,000 users
- Developed the user knowledge graph service using Neo4j, improving the performance of read queries threefold

Software Engineer

Jun 2016 – Mar 2017

Snaappy

Moscow, Russia

- Rewrote the messaging service in Go, which increased the backend performance fivefold
- Migrated the chat storage from PostgreSQL to MongoDB, allowing horizontal scalability for the data layer
- Implemented a WebSocket service for real-time updates using Go, driving user engagement by 30%

TEACHING EXPERIENCE

Graduate Teaching Assistant

Sep 2025 – Dec 2025

NYU Courant Institute of Mathematical Sciences. Prof. Saining Xie

New York, United States

- Led weekly office hours for CSCI-GA.2271-001 Computer Vision. [Website](#)
- Helped students with their research projects and homework

PUBLICATIONS

1. *Solaris: Building a Multiplayer Video World Model in Minecraft*
G. Savva*, O. Michel*, D. Lu*, P. Waiwitlikhit, T. Meehan, D. Mishra, S. Poddar, J. Lu, S. Xie.
Under Review 2026. [Website](#)
2. *HuDOR: Bridging the Human to Robot Dexterity Gap through Object-Oriented Rewards*
I. Guzey, Y. Dai, **G. Savva**, R. Bhirangi, L. Pinto.
ICRA 2025. [Website](#)

PROJECTS

Do Pre-Trained and Fine-Tuned World Models Generalize?

Jan 2025 – May 2025

Investigated the generalization and finetuning of two SOTA Minecraft World Models on three data distributions.
[Website](#)

AppSim: A Learned World Model for an App API

Jan 2025 – May 2025

Used ChatGPT o3 in a zero-shot setting, achieving 74% accuracy, and compared its performance to a finetuned TinyLlama model. [Website](#)

Transformer-Based Diffusion for Game Generation

Oct 2024 – Dec 2024

Trained a transformer diffusion model to simulate DOOM trajectories. It achieves a PSNR of 32.21 in the teacher-forcing setting, producing an indistinguishable quality from the ground truth. [Website](#)

Advantage Actor-Critic with Optuna

Jul 2024 – Sep 2024

Trained a reinforcement learning agent with Optuna for the HalfCheetah env, achieving a 24% better performance than the best publicly available policy. [Website](#)

Open-Source Library Scany

May 2020 – Present

Created an open-source library to map data from a database into Go objects. The library has 1,500 stars on GitHub and is used by thousands of companies. [GitHub](#)

SKILLS

Languages: Python, SQL, Go, TypeScript

Machine Learning: Transformers, Diffusion, U-Net, CNN, LSTM, Reinforcement Learning, Distributed Training, TPU

Libraries: PyTorch, JAX, Gym, Optuna, Hydra, TensorFlow, Numpy, Pandas, FastAPI, React, Node.js

Tools: Docker, Kubernetes, PostgreSQL, AWS, GraphQL, Hasura